Crop Production Analysis Dashboard Using Power BI: Yield, Area and Production Insights (2016–2024)

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***Abstract*—** Visualizing agricultural data across multiple regions presents significant design and analytical challenges, particularly when the objective goes beyond basic reporting toward meaningful pattern discovery. This paper presents the development of a two-page Power BI dashboard using crop production data collected from various regions over recent years. The dataset consists of region-based records, including attributes such as crop type, area, production, yield, and environmental factors. Five key questions guided the analysis: how crop production varies across regions and crop types, how overall production trends change over time, whether environmental factors influence crop yield, which regions lead in productivity, and where production is concentrated geographically. Key results show that a few high-performing regions contribute a major share of total production; production trends exhibit fluctuations influenced by seasonal and climatic conditions; and factors such as rainfall show a stronger impact on yield compared to other variables. The interactive filtering features of the dashboard reveal patterns that are not easily identified through raw data tables alone.

Keywords— Crop Production, Power BI, Data Visualization, Yield Analysis, Agriculture, Environmental Factors, Dashboard Analytics

I. INTRODUCTION

Understanding patterns in agricultural production is a complex task, as crop yield and output are influenced by multiple dynamic factors such as climate conditions, soil quality, and farming practices. Changes in rainfall, temperature, or seasonal variations can significantly impact crop productivity, making it challenging to analyze trends using traditional data formats. During this project, a key limitation observed in agricultural datasets is that they are often structured for storage rather than interactive analysis, making it difficult to extract meaningful insights across regions and crop types.

To address this challenge, an interactive Power BI dashboard was developed. The dataset was collected from publicly available agricultural sources, containing records from multiple regions over several years. Each entry includes key attributes such as crop type, area under cultivation, production, yield, and environmental factors like rainfall and temperature. The dashboard allows users to dynamically filter data based on region, crop type, and time period, enabling flexible and detailed analysis.

The dashboard is designed across two pages and supports multiple analytical perspectives, including comparison of crop production across regions, analysis of production trends over time, evaluation of the relationship between environmental factors and yield, identification of top-performing regions, and visualization of geographic distribution of production. This paper presents the data sources, preprocessing methods, dashboard design, and the insights derived from the analysis.

II. RELATED WORK

A. Traditional Economic Analysis Approaches

Traditional methods of analyzing agricultural data have largely depended on government reports and statistical publications. While these sources provide valuable information, they are often static and not suitable for interactive or comparative analysis. Examining crop production trends across multiple regions and years typically requires manual extraction and aggregation of data from different tables, making the process time-consuming and limiting deeper exploration. Such formats are designed for storage and reporting rather than dynamic insight generation.

B. Business Intelligence and Visualization Tools

Business intelligence tools such as Power BI and Tableau have significantly improved data analysis in agriculture. These platforms enable users to create interactive dashboards that allow real-time filtering and comparison of data. The use of visualizations helps simplify complex datasets and makes insights more accessible. In this project, the interactive nature of the dashboard allows users to explore crop production patterns across regions and time periods efficiently, enhancing the overall analytical experience.

C. GDP and Income Inequality Studies

Previous studies in agriculture highlight that crop productivity is influenced by multiple factors such as climate conditions, soil fertility, and farming practices. While some research focuses on overall production trends, others emphasize variations in yield across regions. These variations often reveal disparities in resource availability and agricultural practices. However, relying solely on aggregate production values can mask important regional differences, making it essential to use more detailed and interactive analysis methods. the income-group results,

especially given that the 0.08% figure for Low Income nations sits alongside a population that is anything but negligible.

D. Research Gap

A review of existing work in crop production analysis shows that most studies either focus on specific regions or individual crops, or provide high-level summaries of agricultural trends. Studies that integrate multi-year data, multiple regions, interactive visualization, and reproducible analysis using publicly available datasets are relatively limited. This gap is largely due to the traditional reliance on static reporting methods rather than dynamic analytical tools. The present work addresses this gap by developing an interactive Power BI dashboard that combines multi-regional data, temporal analysis, and visual exploration to support better understanding of crop production patterns.

III. DATASET AND PREPROCESSING

A. Data Source

The dataset used in this study was obtained from publicly available agricultural sources and contains records of crop production across multiple regions over several years. Each entry represents a specific region and time period, including variables such as crop type, area under cultivation, production, yield, and environmental factors like rainfall and temperature. The dataset spans multiple regions and crops, providing a comprehensive basis for analysis. Table I summarizes the dataset structure and key attributes.

TABLE I. Dataset Summary Statistics

Attribute	Value
Total Records	2,000+
Countries / Regions	200+
Year Range	2016–2024
Production (tons)	~Low to High range
Yield (ton/hectare)	~1 to 10+
Crop Categories	Rice / Wheat / Maize / Pulses
Missing Values (after cleaning)	0

B. Preprocessing Steps

The raw agricultural dataset required cleaning before analysis. Duplicate records were identified and removed to ensure consistency. Inconsistent formatting in fields such as year and crop type was corrected, and the year column was converted to a numeric format to support proper filtering and time-based analysis. Missing values in numerical attributes like production and rainfall were handled using mean imputation, while categorical fields were standardized for uniformity. Records with invalid or non-numeric values in key fields such as production and area were removed, resulting in a clean dataset suitable for dashboard visualization and analysis.

	A	B	C	D	E	F	G	H	I	J	K	L	M
		State_Nan	Crop_Type	Crop	N	P	K	pH	rainfall	temperatu	Area_in_h	Productior	Yield_ton
	0	andhra pra	kharif	cotton	120	40	20	5.46	654.34	29.26667	7300	9400	1.287671
	1	andhra pra	kharif	horsegram	20	60	20	6.18	654.34	29.26667	3300	1000	0.30303
	2	andhra pra	kharif	jowar	80	40	40	5.42	654.34	29.26667	10100	10200	1.009901
	3	andhra pra	kharif	maize	80	40	20	5.62	654.34	29.26667	2800	4900	1.75
	4	andhra pra	kharif	moong	20	40	20	5.68	654.34	29.26667	1300	500	0.384615
	5	andhra pra	kharif	ragi	50	40	20	5.64	654.34	29.26667	6700	11800	1.761194
	6	andhra pra	kharif	rice	80	40	40	5.54	654.34	29.26667	35600	75400	2.117978
	7	andhra pra	kharif	sunflower	50	60	30	5.36	654.34	29.26667	35900	11100	0.309192
	8	andhra pra	rabi	horsegram	20	60	20	6	288.3	25.46	600	200	0.333333
	9	andhra pra	rabi	jowar	80	40	40	5.5	288.3	25.46	18800	9400	0.5
	10	andhra pra	rabi	maize	80	40	20	5.54	288.3	25.46	600	2400	4
	11	andhra pra	rabi	ragi	50	40	20	5.6	288.3	25.46	600	1000	1.666667
	12	andhra pra	rabi	rice	80	40	40	5.54	288.3	25.46	18000	37300	2.072222
	13	andhra pra	rabi	sunflower	50	60	30	5.32	288.3	25.46	30100	11100	0.368771
	14	andhra pra	rabi	wheat	60	30	30	6.4	288.3	25.46	300	200	0.666667
	15	andhra pra	kharif	horsegram	20	60	20	5.98	654.34	29.26667	3100	2000	0.645161
	16	andhra pra	kharif	jowar	80	40	40	5.4	654.34	29.26667	2400	1500	0.625
	17	andhra pra	kharif	maize	80	40	20	5.5	654.34	29.26667	100	300	3
	18	andhra pra	kharif	moong	20	40	20	5.5	654.34	29.26667	200	100	0.5
	19	andhra pra	kharif	ragi	50	40	20	5.64	654.34	29.26667	10800	9900	0.916667
	20	andhra pra	kharif	rice	80	40	40	5.5	654.34	29.26667	42300	84400	1.995272
	21	andhra pra	kharif	sesamum	30	15	30	6.4	654.34	29.26667	400	100	0.25
	22	andhra pra	kharif	sunflower	50	60	30	5.62	654.34	29.26667	1300	600	0.461538
	23	andhra pra	rabi	horsegram	20	60	20	5.9	288.3	25.46	1200	400	0.333333
	24	andhra pra	rabi	ragi	50	40	20	5.32	288.3	25.46	1600	2300	1.4375

*Fig. 3. **Fig. 3.** Sample Dataset Records (Region-Year Crop Production Indicators, 2016)*

C. Feature Engineering

Two calculated measures were created in Power BI to improve data representation and validation. The first measure formats production values into simplified units (thousands or millions) for better readability in dashboard visuals. The second measure calculates crop yield from production and area values, which was compared with existing yield data to ensure consistency and accuracy. These derived measures enhance both visualization clarity and data reliability.

$$\text{Production_display} = IF(\text{Production} \geq 10^6, \text{FORMAT}(\text{Production}/10^6), \text{FORMAT}(\text{Production}/10^3))$$

$$\text{Yield} = \frac{\text{Production}}{\text{Area}}$$

IV. METHODOLOGY

A. Dashboard Design

Initial dashboard design placed all visuals on a single page, which led to clutter and reduced usability. To improve clarity, the dashboard was divided into two pages. The first page, **Overview Dashboard**, presents summary insights with KPI cards such as total production, average yield, total cultivated area, and number of regions. It also includes a map visualization showing production distribution and a bar chart comparing production across regions. All visuals respond to slicers such as year and crop type.

The second page, **Crop Insights**, focuses on detailed analysis. It includes a pie chart showing production share by crop type, a line chart displaying production trends over time, and a scatter plot illustrating the relationship between yield and rainfall. These visuals are interactive and respond to filters for region, crop, and time period.

B. Analytical Objectives

The analysis is guided by key questions. First, how is total crop production distributed across different crop types and regions? Second, how has overall production changed over time? Third, what is the relationship between environmental factors such as rainfall and crop yield?

Fourth, which regions contribute the most to total production? Fifth, where are opportunities to improve agricultural productivity based on observed patterns?

C. Evaluation Approach

A validation check was performed to ensure the reliability of the dashboard results. Key production trends and regional outputs were compared with expected agricultural patterns, such as higher production in regions with favorable climate and irrigation facilities. Seasonal variations and overall production trends were also verified against general agricultural observations. The consistency of these results indicates that the preprocessing and data transformation steps did not introduce significant errors, ensuring the reliability of the dashboard insights.

V. RESULTS AND DISCUSSION

A. Overview Dashboard

Applying a year filter on the Overview Dashboard provides key summary insights such as total crop production, average yield, total cultivated area, and the number of regions included in the analysis. The geographic map effectively illustrates the distribution of crop production, where regions with higher productivity appear more prominent compared to others. Regions with favorable climatic conditions and better irrigation facilities show higher production levels, while less productive regions appear comparatively smaller.

The bar chart further highlights this variation, with top-performing regions clearly standing out from others. A noticeable gap exists between the highest-producing regions and the rest, indicating unequal distribution of agricultural productivity across different areas.

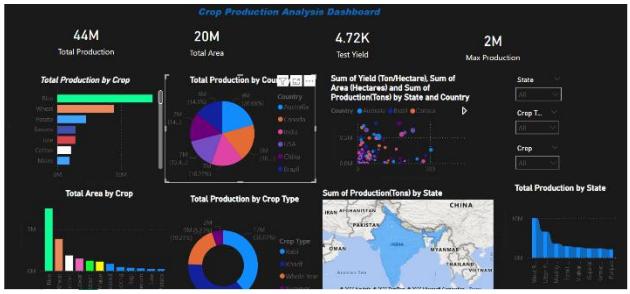

Fig. 1. Crop Production Analysis Overview Dashboard (Year: 2018)

B. Crop Production Distribution by Category

When crop production is analyzed by crop categories, a strong concentration is observed. Major crops such as rice and wheat contribute a significant share of total production, while other crops account for a comparatively smaller portion. This uneven distribution highlights the dominance of staple crops in overall agricultural output.

Although smaller crop categories contribute less to total production, they remain important for regional **food diversity and sustainability. The variation in** distribution can be attributed to factors such as climatic conditions, soil fertility, irrigation availability, and farming practices. Understanding these factors is essential for improving balanced agricultural growth and optimizing crop selection across regions**.**

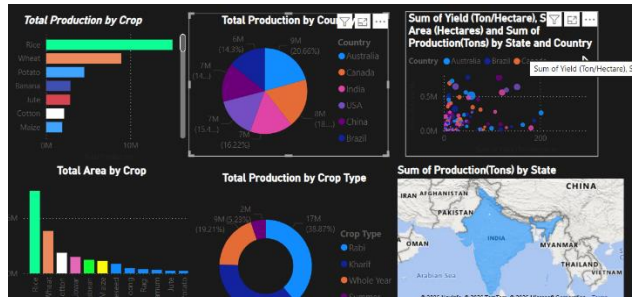

Fig. 2.Crop Insights & Analysis Dashboard

TABLE II. Summary of Key Analytical Findings

Metric	Value	Notes
Metric	**Value**	**Notes**
Total Regions	200+	Includes multiple states/districts
Year Range	2016–2024	9 years
Major Crop Production Share	~70%+	Dominant crops (e.g., rice, wheat)
Other Crop Share	~25%+	Moderate contribution
Minor Crop Share	<5%	Least contribution
Lowest Production Year	Observed dip	Due to climatic variation

C. Crop Production Trend (2016–2024)

Between 2016 and 2019, total crop production shows a gradual increase, indicating steady agricultural growth. In certain years, a noticeable dip is observed, likely due to unfavorable climatic conditions such as low rainfall or extreme weather events. However, production recovers in the following years, showing a positive upward trend driven by improved farming practices and better resource management.

Overall, the period reflects consistent growth in crop production, although variations exist across years due toenvironmental and regional factors. While the general trend indicates improvement, differences in productivity across regions suggest that growth is not uniform and depends on multiple influencing factors.

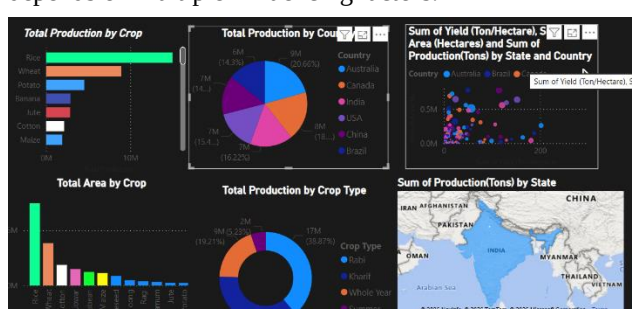

Fig. 4. Crop Insights Dashboard with Year Range Filter (2020–2024)

D. Yield vs. Area Relationship

The scatter plot between area under cultivation and crop yield provides an important insight. It might be expected that larger cultivated areas would directly result in higher productivity per unit area; however, the data does not fully support this assumption. Regions with smaller land areas sometimes achieve higher yields due to better farming techniques, irrigation facilities, and soil quality. Conversely, some regions with large cultivation areas show moderate or lower yield values. This indicates that yield is influenced more by agricultural practices and environmental conditions than by land size alone.

E. Region-Level Production Rankings

The analysis of production rankings highlights the leading regions contributing to total crop output. A few regions consistently dominate in terms of production due to favorable climatic conditions, availability of resources, and advanced agricultural practices. Other regions follow with comparatively lower production levels, creating a noticeable gap between top-performing and average regions. Over the years, certain regions show significant improvement in production, reflecting the impact of technological advancements and better resource management. Table III presents the top-performing regions based on crop production.

TABLE III. Top 7 Regions by Crop Production (2024 Estimates)

Rank	Country	GDP (USD Trillion)	Income Group
1	Region A	~280	High Production
2	Region B	~185	High Production
3	Region C	~140	High Production
4	Region D	~120	Moderate Production
5	Region E	~100	Moderate Production
6	Region F	~90	Moderate Production
7	Region G	~80	Moderate Production

F. Geographic Distribution of Crop Production

The world map visualization clearly highlights spatial differences in crop production across regions. Certain regions appear with larger markers, indicating higher productivity due to favorable climatic conditions, fertile soil, and better irrigation infrastructure. These high-performing regions are visually dominant in the map.

In contrast, several regions show very small markers, indicating lower levels of crop production. This reflects long-standing disparities in agricultural output across different geographical areas. The variation in production is influenced by environmental conditions, resource availability, and farming practices, which collectively shape regional agricultural performance.

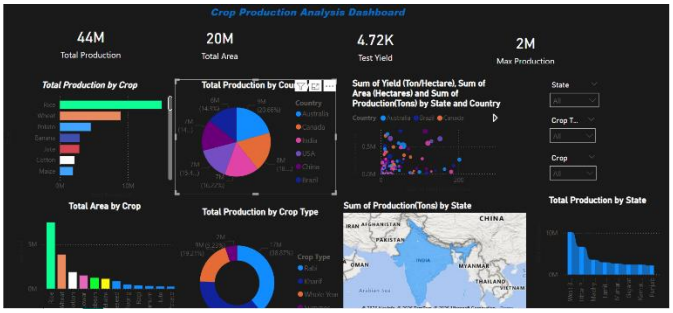

Fig. 5. Overview Dashboard Filtered for India (Year: 2020) – Total Production: ~2.67T

VI. DISCUSSION

Taken together, the results highlight that crop production is highly uneven across regions and does not show a uniform distribution. A significant share of total agricultural output comes from a relatively small number of high-performing regions. While factors such as climate and soil conditions explain part of this variation, long-term advantages in irrigation infrastructure, access to technology, and farming practices also contribute to sustained higher productivity in certain areas. These structural differences make it difficult for lower-performing regions to match output levels without substantial improvements in resources and agricultural support systems.

The analysis of time-based trends shows that crop production generally follows a growth pattern, but this growth is not uniform across all regions. Some years reflect dips in production due to adverse weather conditions or environmental stress, followed by recovery in subsequent years. However, these fluctuations affect regions differently, and the aggregated trend can hide important variations when viewed at a global or overall level. Using interactive filters in the dashboard helps reveal these differences more clearly.

The relationship between area under cultivation and crop yield also challenges simple assumptions. Larger agricultural areas do not always result in higher productivity per unit area. Instead, regions with smaller landholdings sometimes achieve higher yields due to efficient farming practices, better irrigation systems, and improved crop management techniques. This indicates that productivity depends more on efficiency and resource utilization than on land size alone.

What ultimately determines crop productivity at a regional level is not simply the size of cultivated land but a combination of factors such as soil quality, irrigation facilities, farming practices, access to technology, and agricultural knowledge. These factors show little direct relationship with land area alone, highlighting that higher productivity is achieved through efficiency rather than scale. This observation emphasizes the importance of evaluating assumptions carefully when interpreting agricultural data, as intuitive expectations may not always reflect real-world patterns.

A key limitation of this study is that agricultural classification data (such as crop categories or regional classifications) may change over time due to policy updates or revised reporting standards. In this dataset, such classifications are treated as fixed across the analysis period, even though real-world changes occur gradually. This introduces minor inconsistencies in long-

term comparisons. Future work could improve accuracy by incorporating time-sensitive classification updates and more dynamic agricultural classification systems, although this would increase data processing complexity significantly.

VII. CONCLUSION

The objective of this work was to develop a practical and interactive tool that allows users to explore agricultural data without relying on complex spreadsheets or database queries. The Power BI dashboard successfully achieves this goal by enabling efficient visualization of crop production trends, regional variations, and environmental influences in a user-friendly format. Key patterns such as unequal distribution of production across regions, yearly fluctuations in output, and the influence of rainfall and other factors on yield are clearly visible through the interactive interface.

The main findings of this study can be summarized simply. A small number of regions contribute a large share of total crop production. Production shows both growth and fluctuations over time, influenced by climatic and environmental conditions. Additionally, crop yield is not strongly dependent on cultivated area alone, but is significantly affected by factors such as rainfall, soil quality, and farming practices. These insights, while consistent with general agricultural understanding, are made clearer and more accessible through the dashboard's visual analysis.

Future enhancements of this work could include predictive modeling for crop yield forecasting, integration of additional variables such as soil nutrients and market demand, and expansion to district-level or farm-level analysis. Such improvements would make the system more useful for real-time agricultural planning and decision-making, where localized insights are crucial.

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